

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

FAST-NUCES, ISLAMABAD CAMPUS

Assignment 01: Artificial Intelligence

Instructor:

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Student Details:

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The Lore: A Scholar's Burden

Greetings, **Lexicon Engineer**. You have been summoned by the *Order of the Infinite Margin*. The Head Librarian has noted a disturbance in the **Silent Scriptorium**—a sentient, logic-defying archive where the floor tiles shift like wet ink.

Your mission is to program an **Automated Ink-Stalker** (your Python agent) to retrieve the *Codex of Non-Euclidean Python* from the depths of the Scriptorium and return it to the Alchemist's Lectern.

1. The Scriptorium Map

The Scriptorium is a 10×10 coordinate plane. Each cell type imposes a specific "Cognitive Load" (Movement Cost).

- **Polished Stone (.)**: The standard path. **Cost: 1**
- **Ink Spills (~)**: Viscous and mentally draining. **Cost: 4**
- **Ancient Desks (&)**: Requires climbing over heavy mahogany. **Cost: 2**
- **Void Pillars (#)**: Structural impossibilities. **Impassable.**
- **S (0,0)**: The Entry Portal (Initial State).
- **G (9,9)**: The Alchemist's Lectern (Goal State).

Official Map of the Silent Scriptorium

R/C	0	1	2	3	4	5	6	7	8	9
0	S	.	.	.	#	.	.	.	.	.
1	.	~	~	.	#	.	&	&	.	.
2	.	~	.	.	.	.	#	&	.	.
3	.	.	.	#	#	.	.	.	.	.
4	#	#	.	.	.	.	.	.	#	#
5	.	.	.	.	~	~	.	.	.	.
6	.	&	&	.	~	.	.	#	#	.
7	.	&	.	.	.	.	.	.	~	.
8	.	.	.	#	#	.	~	~	~	.
9	.	.	.	.	#	.	.	.	.	G

2. The Scholarly Constraints

Standard pathfinding will fail here. Your agent must adhere to the **Laws of the Order**:

1. **The Claustrophobia Paradox:** A scholar's mind requires space to breathe. You **cannot** enter any cell (r, c) if that cell is adjacent (up, down, left, right) to **more than two** Void Pillars (#).
2. **The Ink-Stained Path:** If you traverse more than **two consecutive** Ink Spill (~) cells, the cost of the current move and all subsequent moves increases by **+2**. This penalty is only removed once you land on a Polished Stone (.) cell.

3. Computation Tasks

Implement the following in Python:

- **Uniform Cost Search (UCS):** Find the mathematically cheapest path.
- **A* Search:** Use **Manhattan Distance** as your heuristic.
- **Beam Search:** Use a **Beam Width** ($W = 3$).